7 A sequence is defined by the recurrence relation

$u_{k+1} = u_k + 5$ with $u_1 = -2$.

(a) Write down the values of u_2, u_3, and u_4. **[1]**

(b) Explain whether this sequence is divergent or convergent. **[1]**

(c) Determine the value of u_{30}. **[2]**

(d) Determine the value of $\sum_{k=1}^{30} u_k$. **[2]**

7(a)

$u_2 =$

$u_3 =$

$u_4 =$

7(b)

7(c)

7(d)